PROJECT REPORT

MACHINE LEARNING

WITH DECISION TREE AND RANDOM FOREST

on

Diamond Cut Classification Using Decision Tree and Random Forest



Submitted by

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Project Domain	:	Machine Learning and Data Science
Programming Language	:	Python
Tools and Technologies	:	Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Jupyter Notebook

1. INTRODUCTION

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions. It is widely used in areas such as healthcare, finance, business analytics, and classification problems.

The project **"Diamond Cut Classification Using Decision Tree and Random Forest"** focuses on applying machine-learning classification algorithms to predict the cut quality of diamonds. The project uses diamond-related features such as carat, depth, table, price, and physical dimensions, along with encoded color and clarity information.

Two machine-learning algorithms, Decision Tree and Random Forest, are implemented and evaluated. Their accuracy scores are compared to understand which model performs better on the test dataset.

2. PROBLEM STATEMENT

Diamond cut quality is an important factor in evaluating a diamond. Identifying cut categories manually or through fixed rules can be challenging when multiple diamond characteristics are involved.

The objective of this project is to use machine-learning techniques to classify diamonds into five cut categories: Fair, Good, Very Good, Premium, and Ideal. The performance of Decision Tree and Random Forest models is compared to determine which provides better classification results on the selected dataset.

3. PROJECT OBJECTIVES

The main objectives of this project are:

- To explore and understand the diamond dataset.
- To perform data preprocessing and encode categorical features.
- To build a Decision Tree classification model.
- To build a Random Forest classification model.
- To evaluate both models using classification performance metrics.
- To compare the models and identify the better-performing approach.
- To visualize model performance and feature importance.

4. DATASET DESCRIPTION

This project uses the Seaborn diamonds dataset, which contains 53,940 records and 10 original columns.

The dataset includes information about diamond characteristics, including carat, cut, color, clarity, depth, table, price, and physical dimensions.

The target variable is **cut**, which contains five categories:

- Fair
- Good
- Very Good
- Premium
- Ideal

The cut column is used as the target for classification. The remaining selected features are used by the machine-learning models to predict the cut category.

5. TOOLS AND TECHNOLOGIES

The following tools and libraries are used in this project:

- **Python:** Main programming language.
- **Pandas:** Data handling and manipulation.
- **NumPy:** Numerical operations.
- **Matplotlib:** Data visualization.
- **Seaborn:** Statistical visualization and dataset exploration.
- **Scikit-learn:** Data preprocessing, model training, and evaluation.
- **Jupyter Notebook:** Writing and executing Python code.

6. METHODOLOGY

The project follows a structured machine-learning workflow.

1. **Dataset loading:** Load the diamonds dataset into a Pandas DataFrame.
2. **Data exploration:** Examine the dataset structure, feature types, and target distribution.
3. **Data preprocessing:** Encode categorical variables into numerical values.
4. **Feature selection:** Select the input features and the target variable.
5. **Train-test split:** Divide the data into training and testing sets using an 80:20 ratio and stratification.

6. **Model training:** Train Decision Tree and Random Forest classifiers.
7. **Model evaluation:** Calculate accuracy and examine confusion matrices.
8. **Comparison:** Compare the results and analyze feature importance.

7. DATA EXPLORATION AND PREPROCESSING

Data exploration helps understand the structure and characteristics of the dataset before model training. The dataset is inspected to identify the available columns, data types, and distribution of diamond cut categories.

The categorical features, such as color and clarity, are converted into numerical representations using label encoding. The target cut categories are also encoded for model training.

The selected input features are carat, depth, table, price, x, y, z, encoded color, and encoded clarity. The cut column is used as the target variable.

The data is divided into training and testing sets. The training set contains 43,152 records, while the testing set contains 10,788 records.

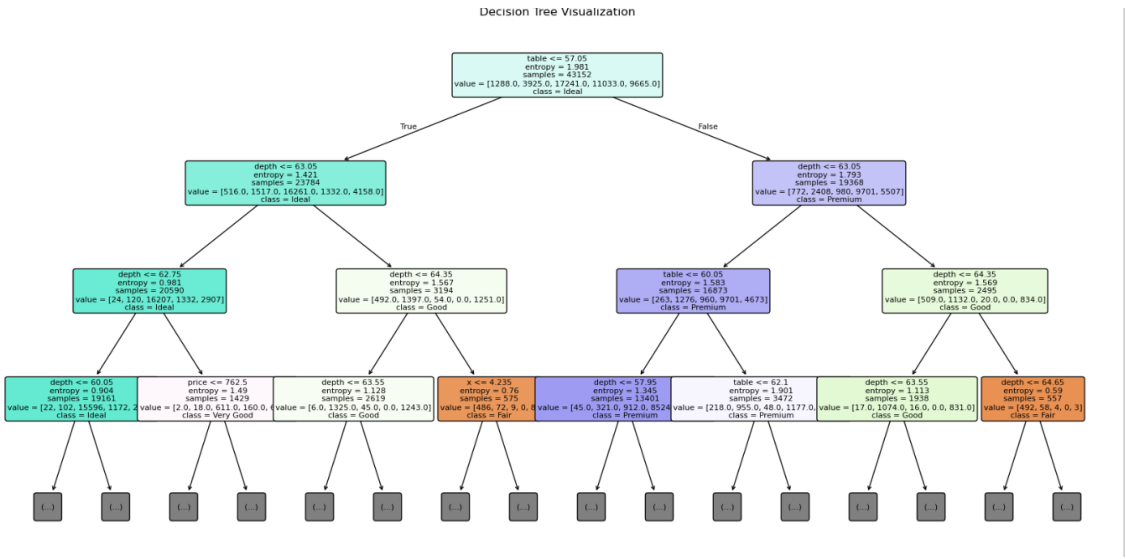
8. DECISION TREE CLASSIFIER

A Decision Tree is a supervised machine-learning algorithm that makes predictions by splitting data into branches based on feature conditions. Each internal node represents a decision, while the leaf nodes represent the predicted class.

In this project, a Decision Tree classifier is trained using the entropy criterion, with a maximum depth of 8 and a random state of 0.

The model learns patterns from the training data and predicts diamond cut categories for the test data. Its performance is evaluated by comparing the predicted categories with the actual categories.

Test accuracy: 74.50%



9. RANDOM FOREST CLASSIFIER

Random Forest is an ensemble-learning algorithm that combines multiple decision trees to produce a prediction. For classification, the trees contribute predictions that are combined to determine the final class.

In this project, a Random Forest classifier is trained using 100 decision trees and the entropy criterion. The random state is set to 0, and parallel processing is enabled.

The model is evaluated on the same test dataset used for the Decision Tree model, allowing a fair comparison of their results.

Test accuracy: 78.59%

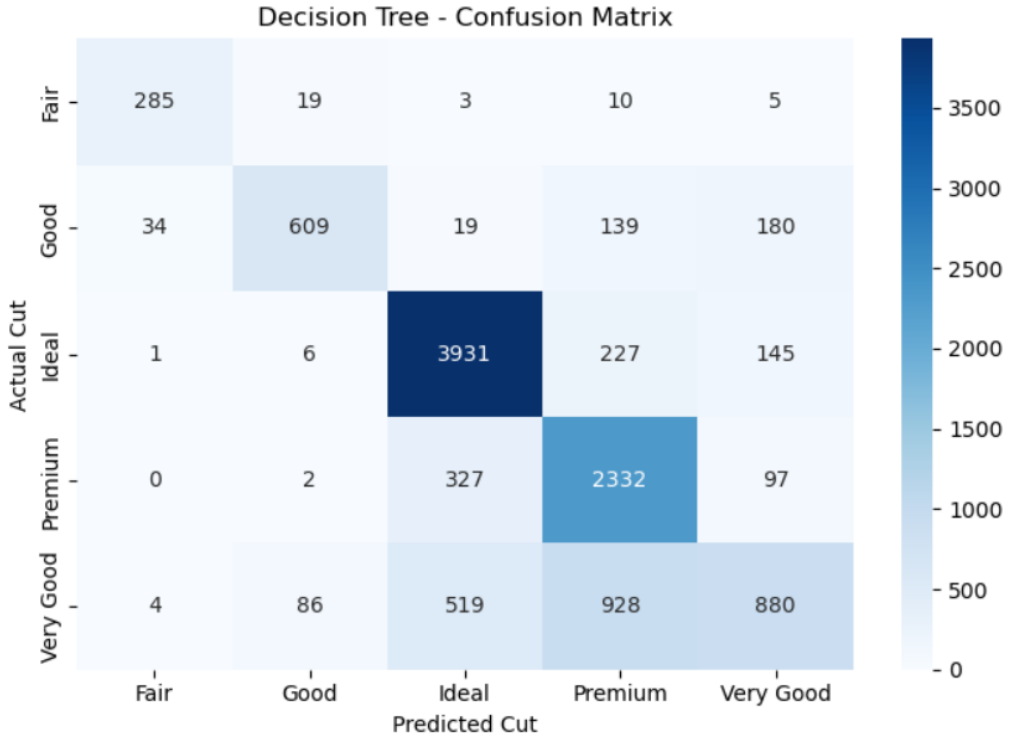
10. MODEL EVALUATION

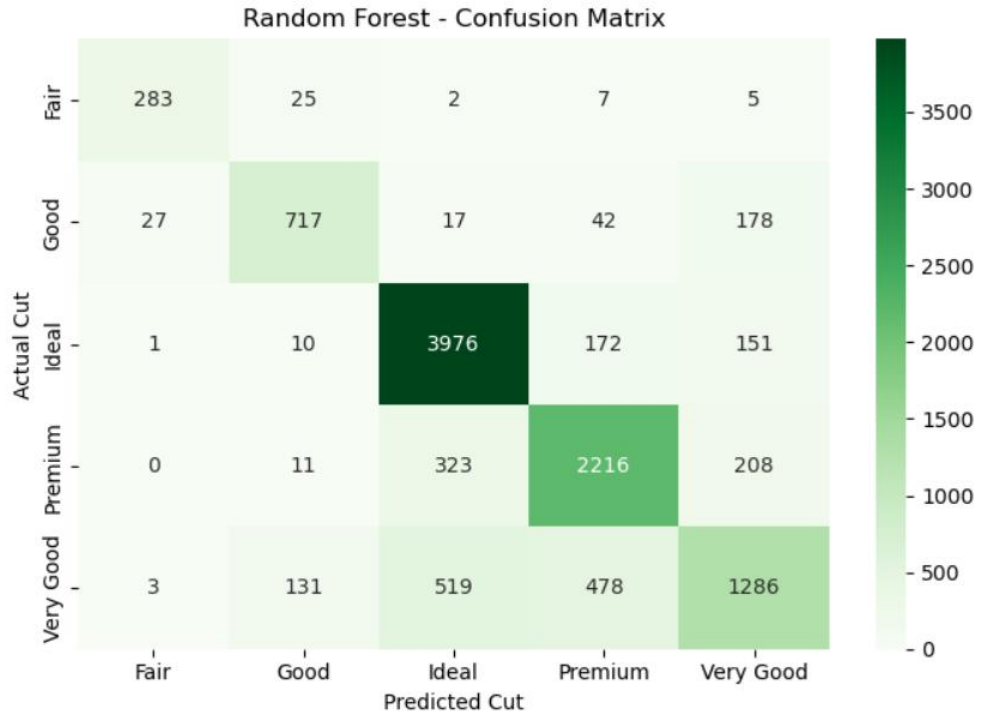
Model evaluation helps determine how effectively the classifiers predict diamond cut categories.

The following evaluation methods are used:

- **Accuracy:** Measures the proportion of correct predictions among all test predictions.
- **Confusion matrix:** Shows correct and incorrect predictions for each cut category.
- **Feature importance:** Indicates how much each input feature contributes to the Random Forest model's decisions.

Accuracy is useful for an overall comparison, while the confusion matrix helps identify categories that the models classify correctly or confuse with other categories.





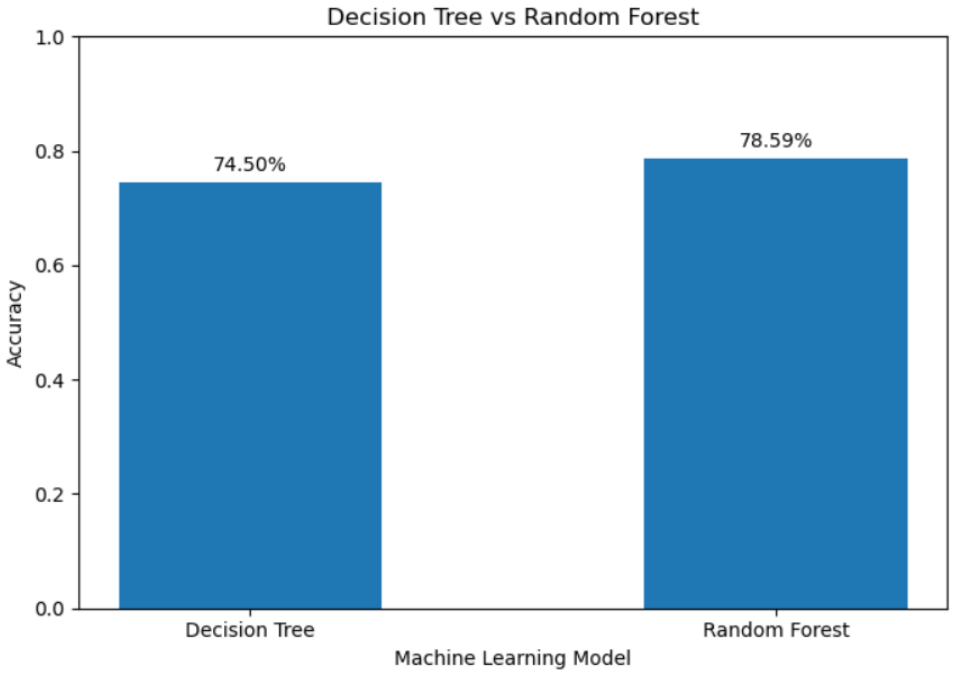
11. RESULTS AND PERFORMANCE COMPARISON

The two trained models produce the following test results:

Model	Accuracy
Decision Tree	74.50%
Random Forest	78.59%

The Random Forest model achieves an accuracy of approximately 78.59%, compared with 74.50% for the Decision Tree model. This is an improvement of approximately 4.09 percentage points.

Based on accuracy, Random Forest performs better on the selected test dataset. However, accuracy alone does not describe performance across every cut category, so the confusion matrices should also be examined.



12. FEATURE IMPORTANCE

Feature importance provides an indication of which features contribute most to the Random Forest model's predictions.

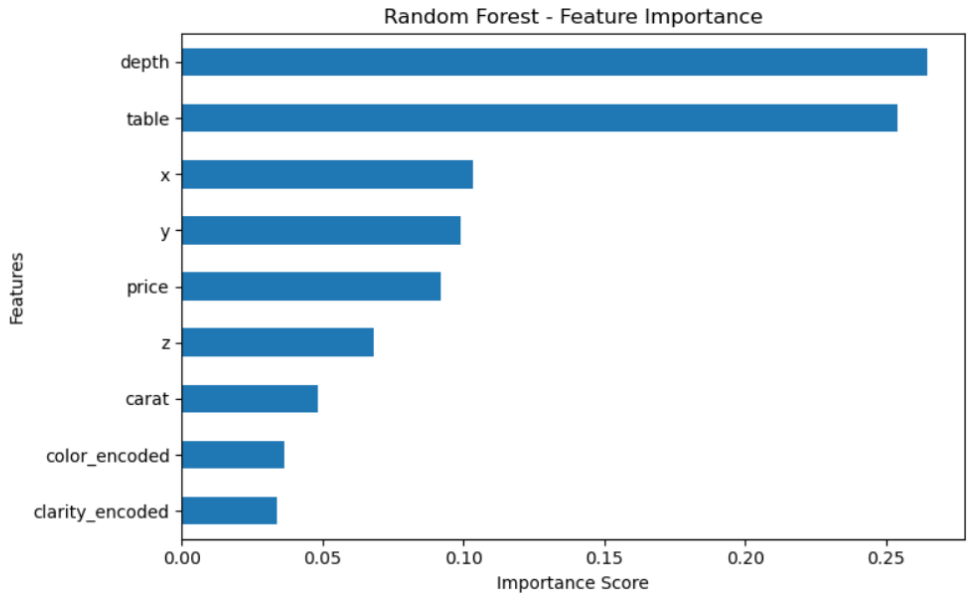
In this experiment, the highest feature-importance values are associated with depth and table, followed by the physical dimensions x and y, and price.

The importance values reported by the model are:

- Depth: 0.2647
- Table: 0.2540
- x: 0.1036

- y: 0.0990
- Price: 0.0920
- z: 0.0683
- Carat: 0.0484
- Color (encoded): 0.0364
- Clarity (encoded): 0.0336

These values describe the model's learned importance scores. They do not prove that a feature independently causes a particular cut quality.



13. LIMITATIONS AND FUTURE SCOPE

Limitations

- The models are evaluated using one train-test split.
- Accuracy may not fully represent performance across all cut categories.
- The dataset contains an unequal number of records across the five target categories.
- Label encoding assigns numerical codes to categorical values, which may influence how some models interpret those features.
- The results depend on the dataset and the selected model settings.

Future Scope

- Evaluate additional classification algorithms.

- Use cross-validation for more reliable performance estimates.
- Tune model hyperparameters to improve results.
- Examine precision, recall, F1-score, and per-class performance.
- Explore alternative encoding methods for categorical features.
- Develop an interactive application that accepts diamond features and displays a predicted cut category.

14. CONCLUSION

This project demonstrates the use of Decision Tree and Random Forest algorithms for diamond cut classification. The Seaborn diamonds dataset is explored and preprocessed before training and testing both models.

The Decision Tree classifier achieves 74.50% accuracy, while the Random Forest classifier achieves 78.59% accuracy. Therefore, Random Forest performs better in terms of overall test accuracy in this experiment.

The project provides practical experience in data preprocessing, supervised machine learning, model evaluation, and visualization using Python and Scikit-learn. Further improvements can be explored through cross-validation, hyperparameter tuning, and detailed per-class evaluation.